Carrito de compras en Angular y Spring Boot + MySQL

Profesor: SinFloo

Fecha: 30/03/2026

- Carrito de compras en Angular y Spring Boot + MySQL - Parte 01 ✅
- Carrito de compras en Angular y Spring Boot + MySQL - Parte 02 ✅
- Carrito de compras en Angular y Spring Boot + MySQL - Parte 03 📌

1 Objetivos

1. Construir carrito de compras
    a. Comprender la arquitectura de una aplicación web Full Stack
    b. Implementar un carrito de compras funcional
    c. Diseñar el modelo relacional de base de datos
    d. Implementar y administrar la base de datos en MySQL
    e. Crear e integrar servicios RESTful (API)
    f. Implementar autenticación y seguridad con JWT y Spring Security
    g. Desplegar la aplicación en un entorno local y en la nube.
    h. Gestionar versiones de código con Git y GitHub

2 Software /herramientas instaladas

1. Visual Studio Code
2. Spring Tools Suite
3. Mysql Workbench 8.0
4. Git /Github

3 Paqueterías instaladas

1. Visual Studio Code - VSC
    a.
2. Spring Boot
    a.
3. Angular
    a.

4 Arquitectura

5 Ambientación - MYSQL

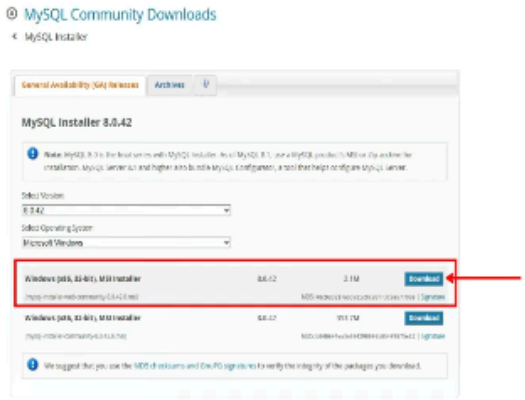
Instalación y ambientación

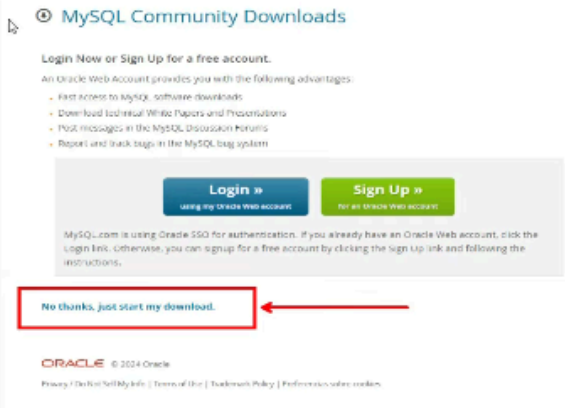
1. Instalación de Mysql.
    a. descargar mysql

➔ Descargar MySQL

- https://dev.mysql.com/downloads/installer/

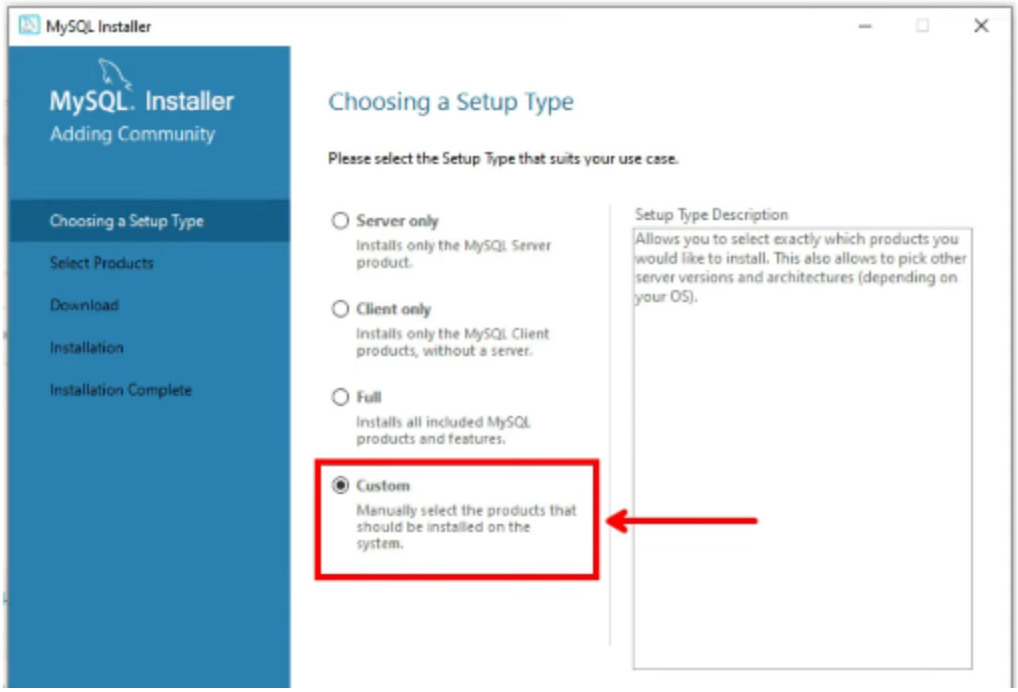
➔

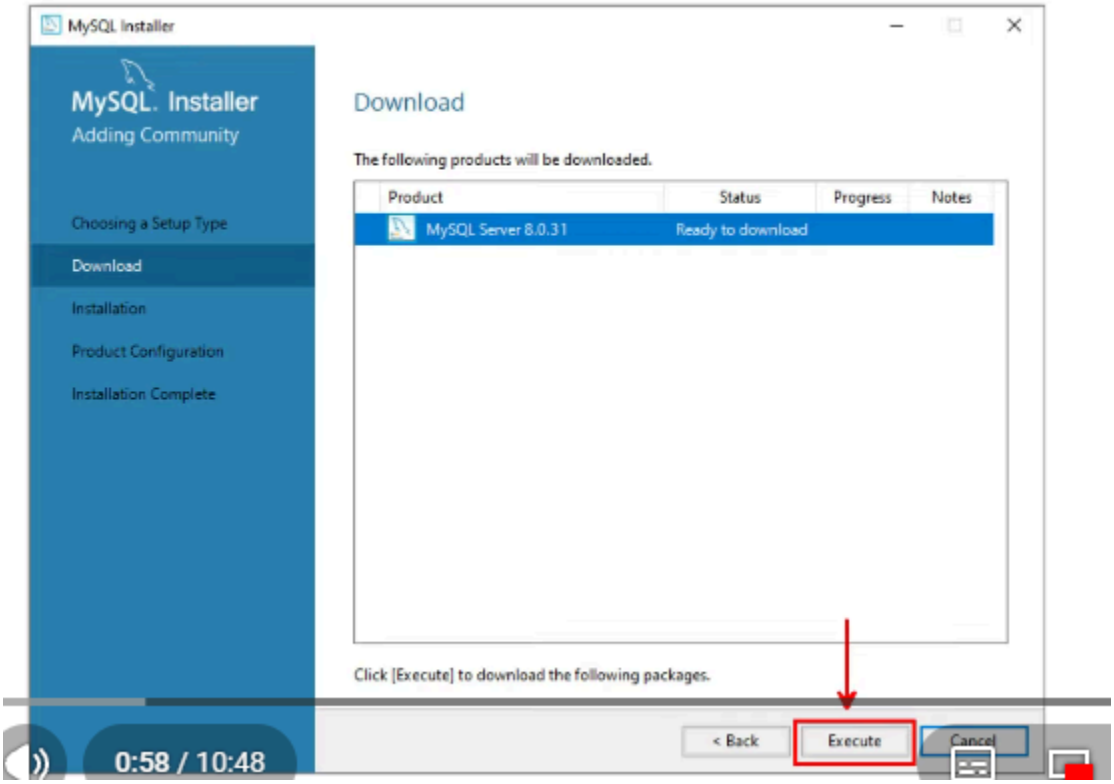


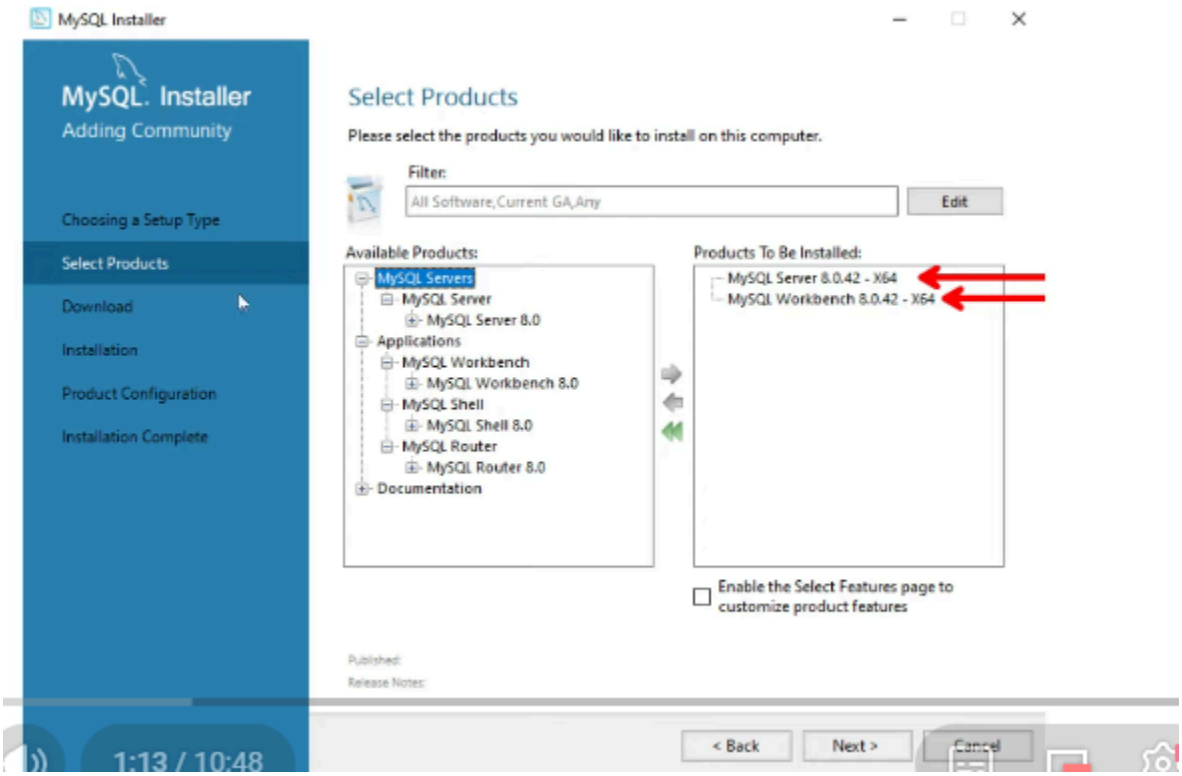


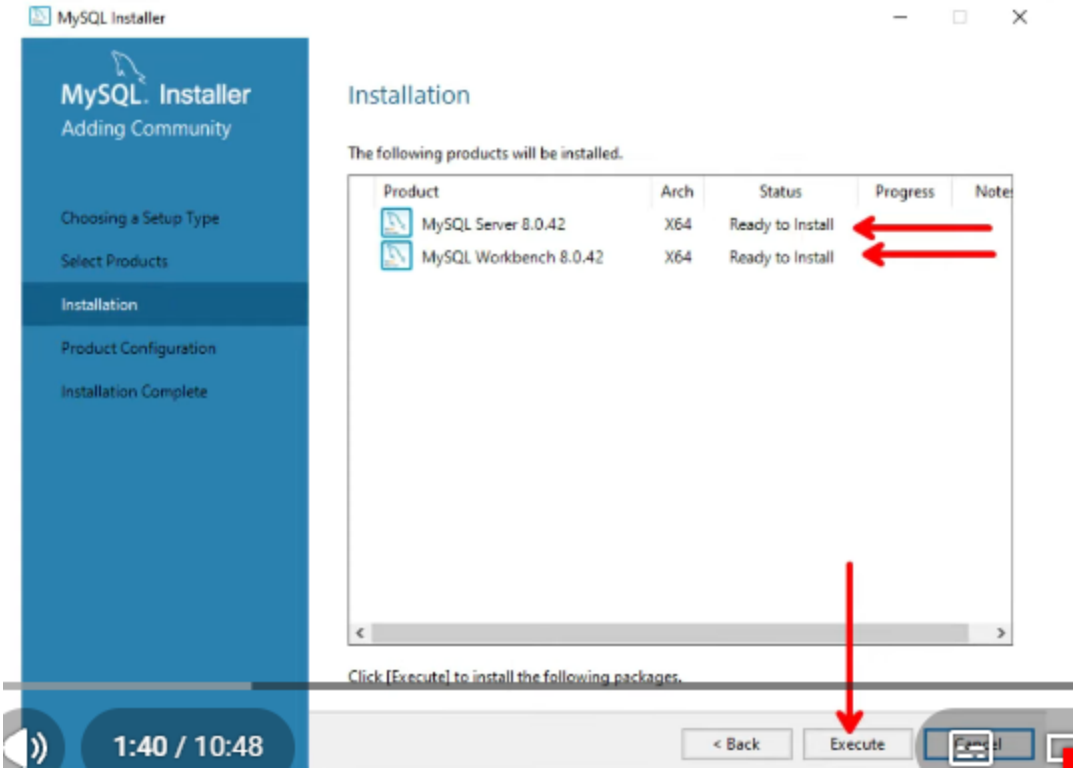
    b. ejecutar .exe

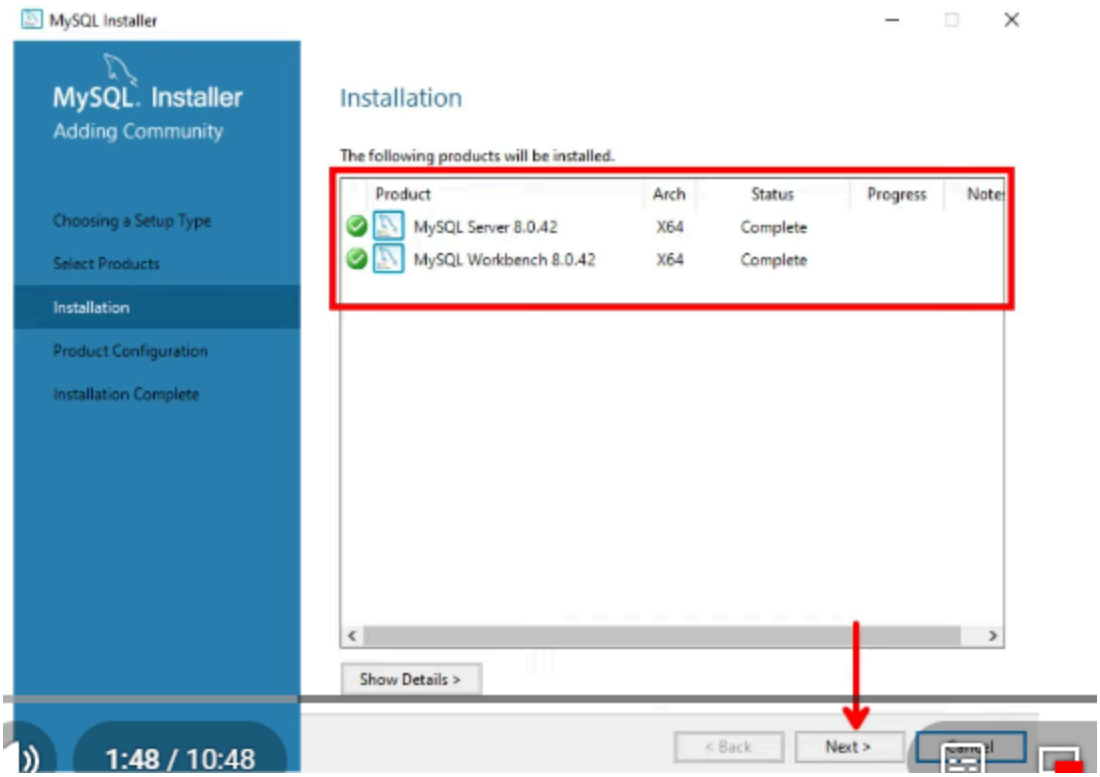
Nombre
Fecha de modificación
Tipo
Tamaño
images
15/06/2025 21:52
Carpeta de archivos
mysql-installer-web-community-8.0.42.0.msi
16/06/2025 17:04
Paquete de Windo...
2,196 KB


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Select Products
Download
Installation
Installation Complete
Choosing a Setup Type
Please select the Setup Type that suits your use case.
Server only
Installs only the MySQL Server product.
Client only
Installs only the MySQL Client products, without a server.
Full
Installs all included MySQL products and features.
Custom
Manually select the products that should be installed on the system.
Setup Type Description
Allows you to select exactly which products you would like to install. This also allows to pick other server versions and architectures (depending on your OS).

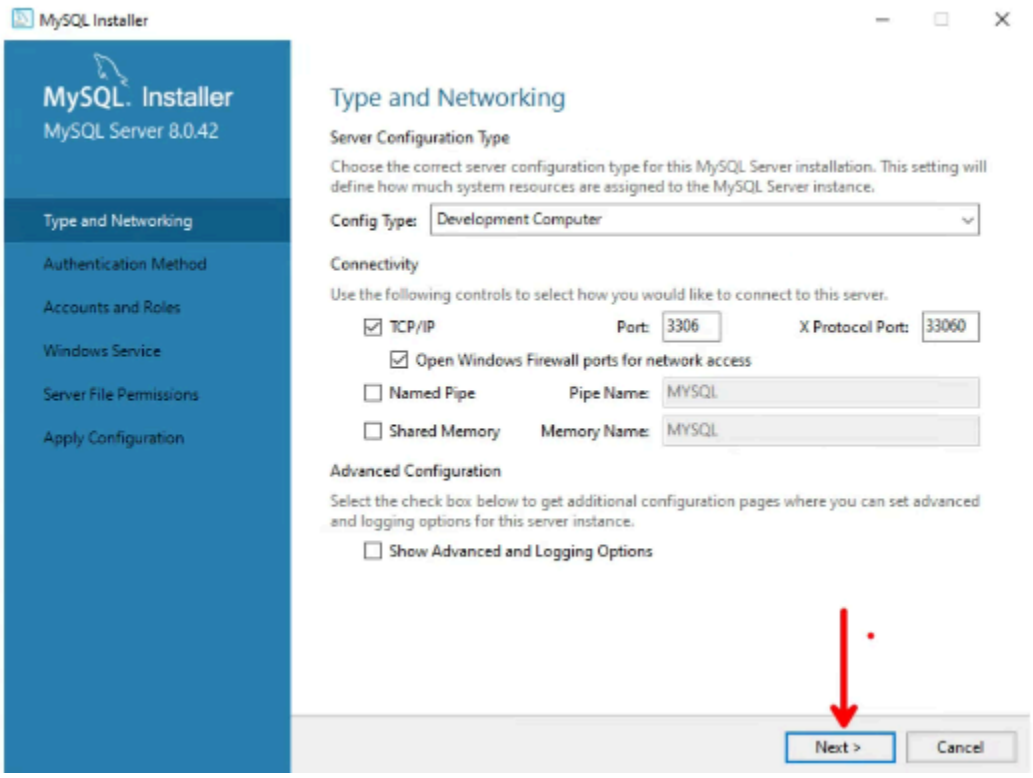

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Download
Installation
Product Configuration
Installation Complete
Download
The following products will be downloaded.
Product
Status
Progress
Notes
MySQL Server 8.0.31
Ready to download
Click [Execute] to download the following packages.
< Back
Execute
Cancel
0:58 / 10:48

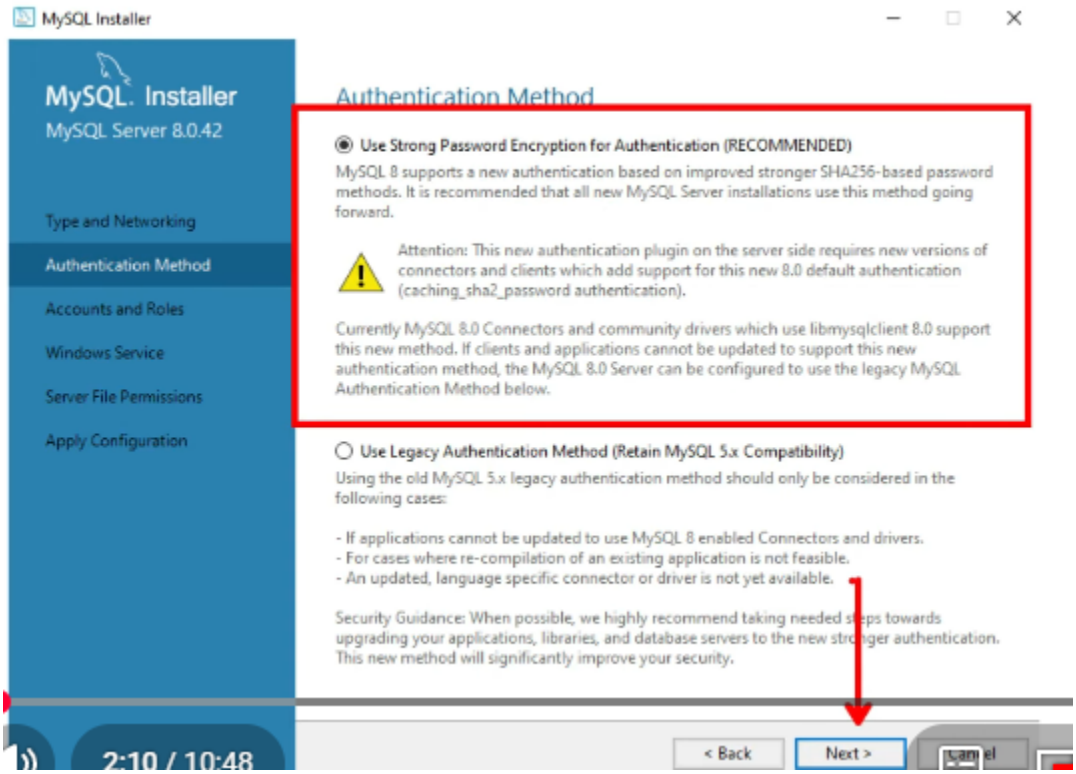

MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Select Products
Download
Installation
Product Configuration
Installation Complete
Select Products
Please select the products you would like to install on this computer.
Filter:
All Software,Current GA,Any
Edit
Available Products:
MySQL Servers
MySQL Server
MySQL Server 8.0
Applications
MySQL Workbench
MySQL Workbench 8.0
MySQL Shell
MySQL Shell 8.0
MySQL Router
MySQL Router 8.0
Documentation
Products To Be Installed:
MySQL Server 8.0.42 - X64
MySQL Workbench 8.0.42 - X64
Enable the Select Features page to customize product features
Published:
Release Notes:
1:13 / 10:48
< Back
Next >
Cancel


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Select Products
Installation
Product Configuration
Installation Complete
Installation
The following products will be installed.
Product
Arch
Status
Progress
Note
MySQL Server 8.0.42
X64
Ready to Install
MySQL Workbench 8.0.42
X64
Ready to Install
Click [Execute] to install the following packages.
1:40 / 10:48
< Back
Execute



lo dejamos por defecto:




MySQL Installer
MySQL. Installer
MySQL Server 8.0.42
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Authentication Method
Use Strong Password Encryption for Authentication (RECOMMENDED)
MySQL 8 supports a new authentication based on improved stronger SHA256-based password methods. It is recommended that all new MySQL Server installations use this method going forward.
Attention: This new authentication plugin on the server side requires new versions of connectors and clients which add support for this new 8.0 default authentication (caching_sha2_password authentication).
Currently MySQL 8.0 Connectors and community drivers which use libmysqlclient 8.0 support this new method. If clients and applications cannot be updated to support this new authentication method, the MySQL 8.0 Server can be configured to use the legacy MySQL Authentication Method below.
Use Legacy Authentication Method (Retain MySQL 5.x Compatibility)
Using the old MySQL 5.x legacy authentication method should only be considered in the following cases:
- If applications cannot be updated to use MySQL 8 enabled Connectors and drivers.
- For cases where re-compilation of an existing application is not feasible.
- An updated, language specific connector or driver is not yet available.
Security Guidance: When possible, we highly recommend taking needed steps towards upgrading your applications, libraries, and database servers to the new stronger authentication. This new method will significantly improve your security.
2:10 / 10:48
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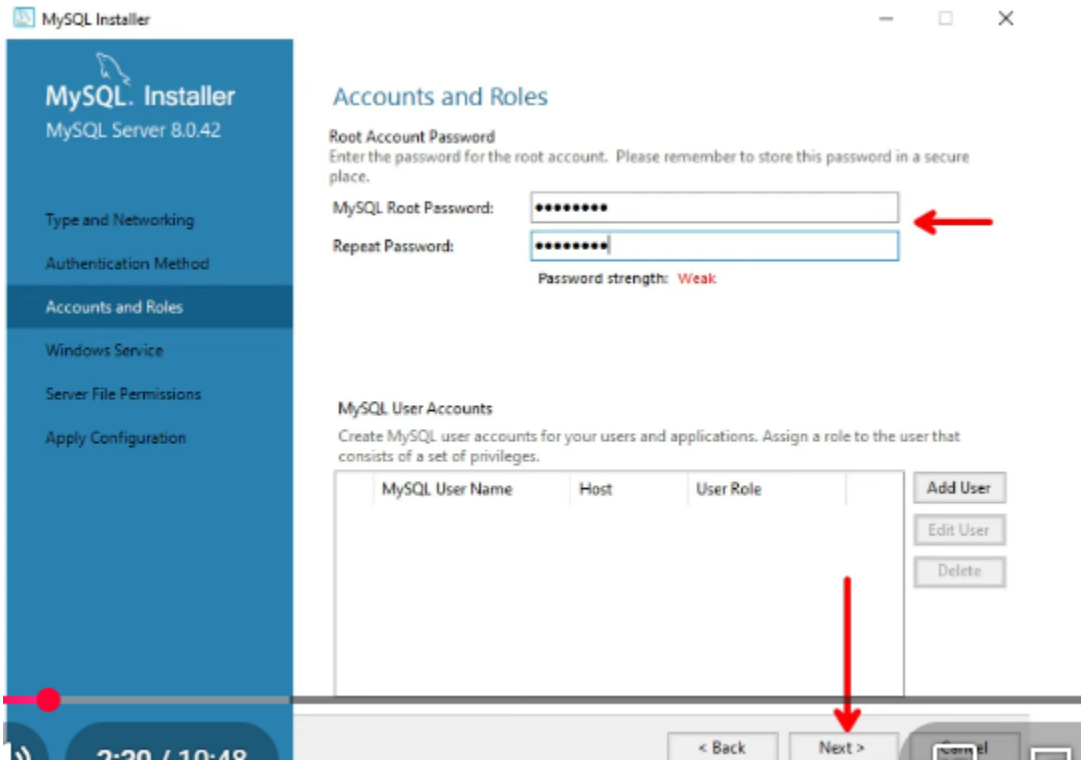
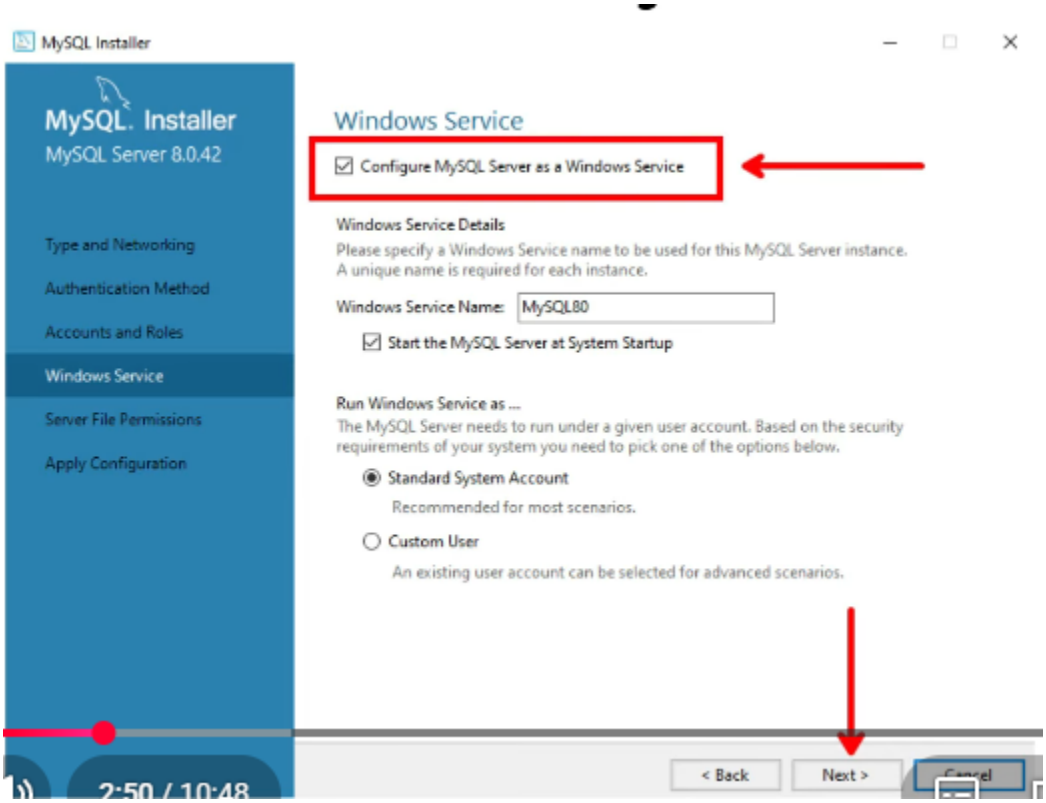
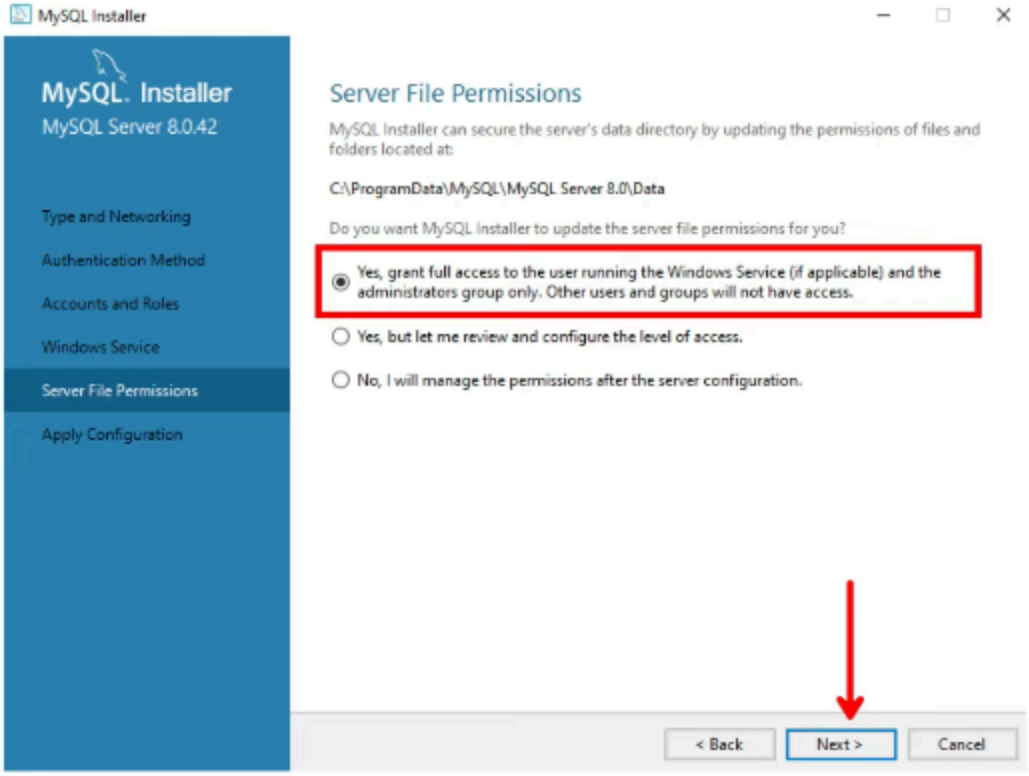
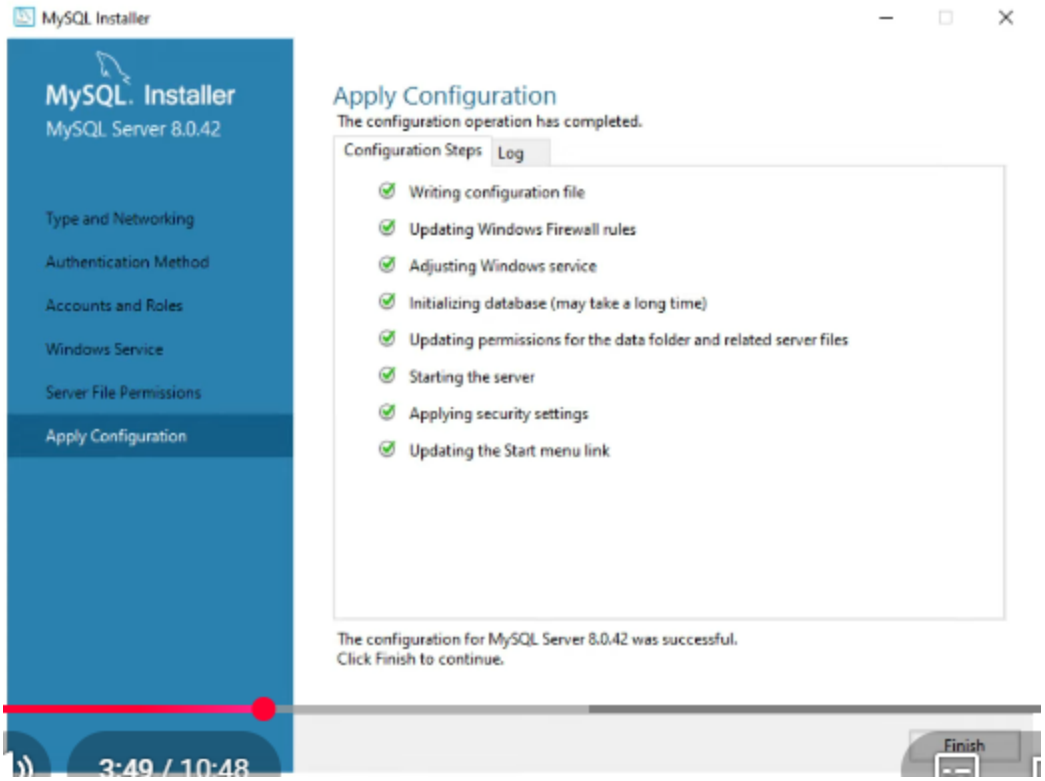
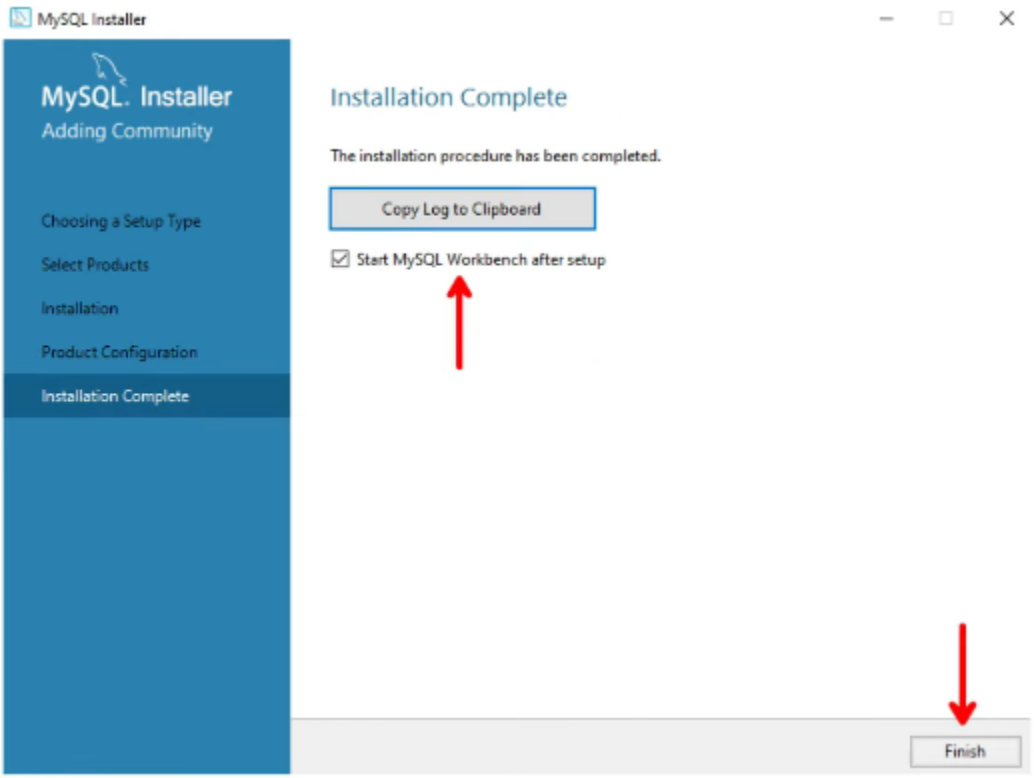
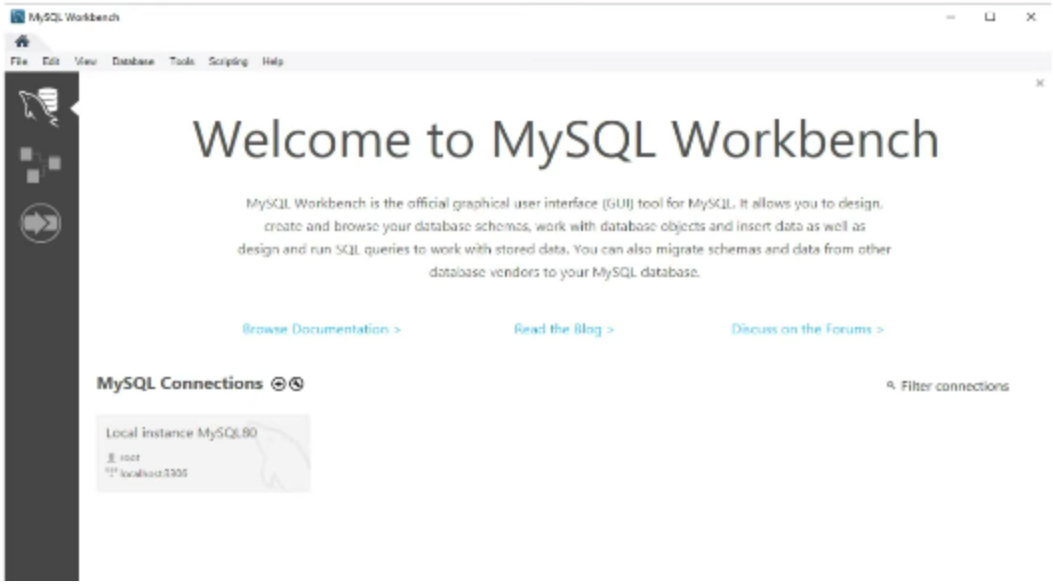

MySQL Installer
MySQL. Installer
MySQL Server 8.0.42
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Accounts and Roles
Root Account Password
Enter the password for the root account. Please remember to store this password in a secure place.
MySQL Root Password:
Repeat Password:
Password strength: Weak
MySQL User Accounts
Create MySQL user accounts for your users and applications. Assign a role to the user that consists of a set of privileges.
MySQL User Name
Host
User Role
Add User
Edit User
Delete
< Back
Next >


MySQL Installer
MySQL. Installer
MySQL Server 8.0.42
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Windows Service
Configure MySQL Server as a Windows Service
Windows Service Details
Please specify a Windows Service name to be used for this MySQL Server instance.
A unique name is required for each instance.
Windows Service Name: MySQL80
Start the MySQL Server at System Startup
Run Windows Service as ...
The MySQL Server needs to run under a given user account. Based on the security requirements of your system you need to pick one of the options below.
Standard System Account
Recommended for most scenarios.
Custom User
An existing user account can be selected for advanced scenarios.
2:50 / 10:48
< Back
Next >
Cancel

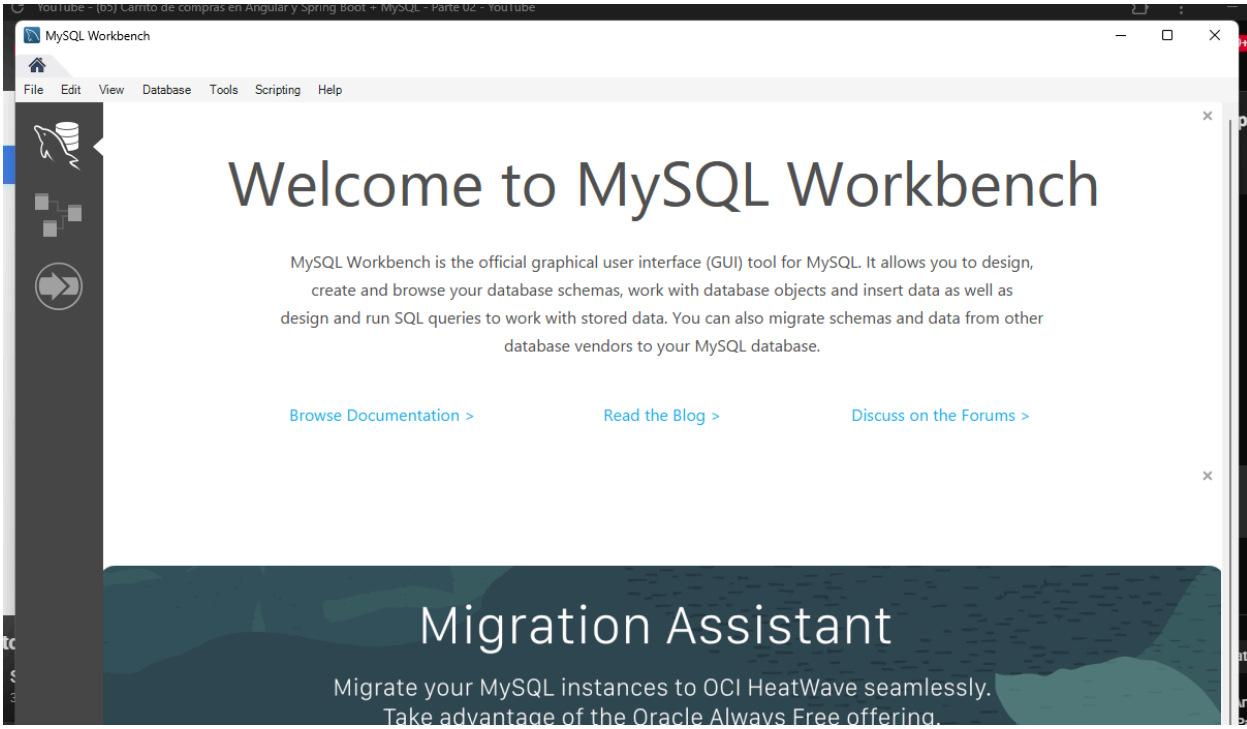

MySQL Installer
MySQL. Installer
MySQL Server 8.0.42
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Server File Permissions
MySQL Installer can secure the server's data directory by updating the permissions of files and folders located at:
C:\ProgramData\MySQL\MySQL Server 8.0\Data
Do you want MySQL Installer to update the server file permissions for you?
Yes, grant full access to the user running the Windows Service (if applicable) and the administrators group only. Other users and groups will not have access.
Yes, but let me review and configure the level of access.
No, I will manage the permissions after the server configuration.
< Back
Next >
Cancel

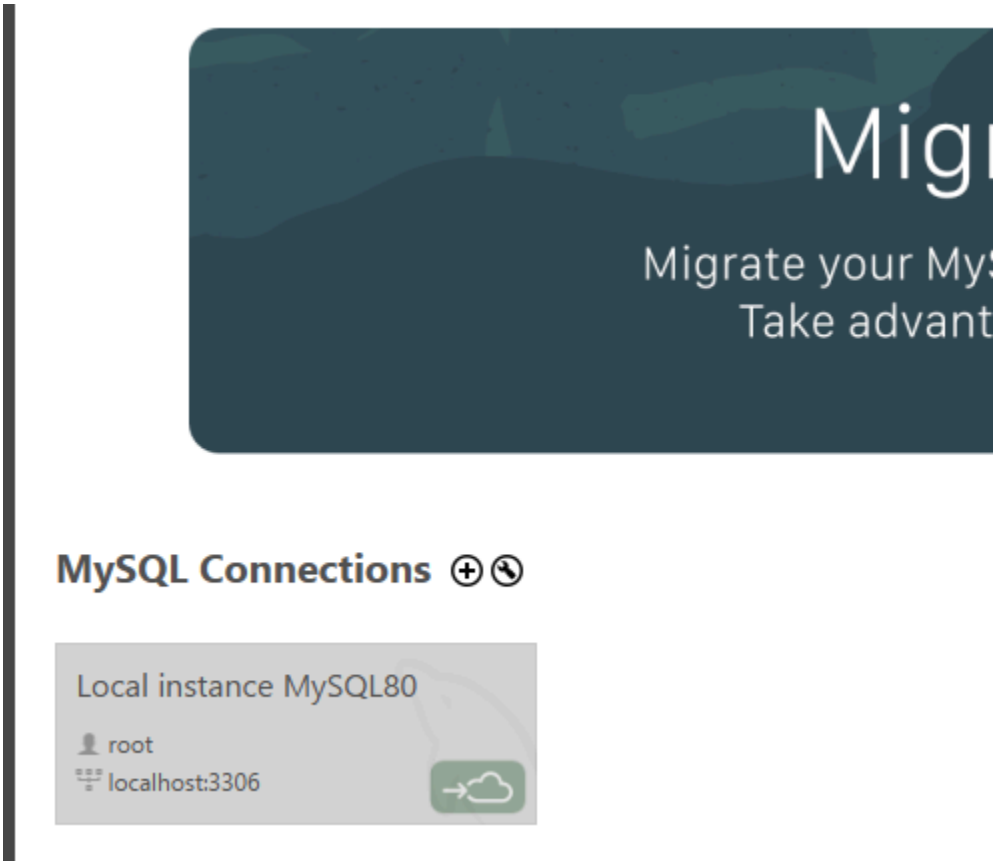

MySQL Installer
MySQL. Installer
MySQL Server 8.0.42
Type and Networking
Authentication Method
Accounts and Roles
Windows Service
Server File Permissions
Apply Configuration
Apply Configuration
The configuration operation has completed.
Configuration Steps
Log
Writing configuration file
Updating Windows Firewall rules
Adjusting Windows service
Initializing database (may take a long time)
Updating permissions for the data folder and related server files
Starting the server
Applying security settings
Updating the Start menu link
The configuration for MySQL Server 8.0.42 was successful.
Click Finish to continue.
3:49 / 10:48
Finish


MySQL Installer
MySQL. Installer
Adding Community
Choosing a Setup Type
Select Products
Installation
Product Configuration
Installation Complete
Installation Complete
The installation procedure has been completed.
Copy Log to Clipboard
Start MySQL Workbench after setup
Finish

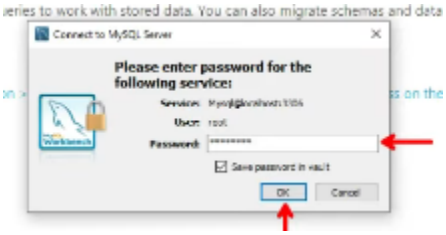

MySQL Workbench
File Edit View Database Tools Scripting Help
Welcome to MySQL Workbench
MySQL Workbench is the official graphical user interface (GUI) tool for MySQL. It allows you to design, create and browse your database schemas, work with database objects and insert data as well as design and run SQL queries to work with stored data. You can also migrate schemas and data from other database vendors to your MySQL database.
Browse Documentation >
Read the Blog >
Discuss on the Forums >
MySQL Connections
Filter connections
Local instance MySQL80
root

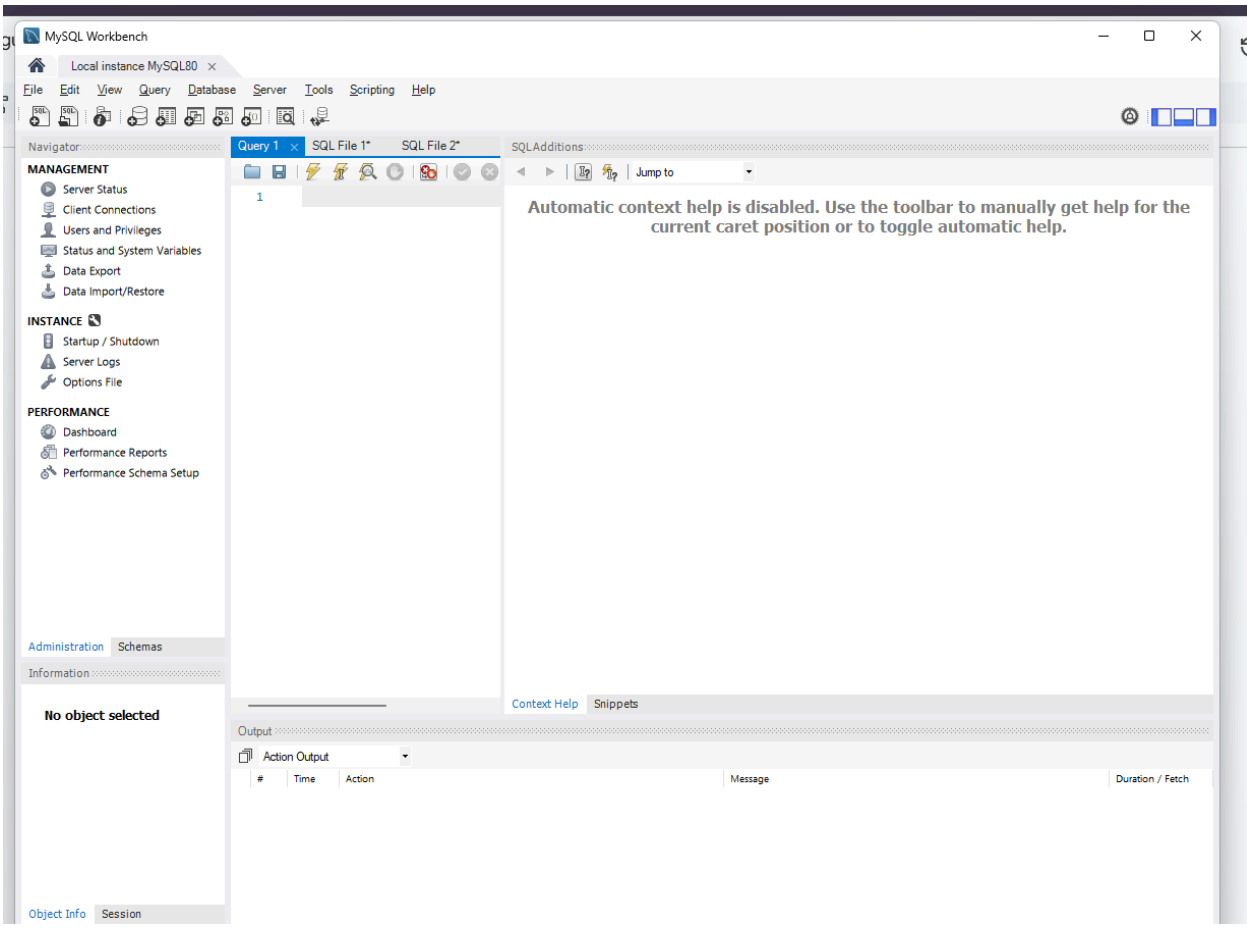
Pantalla de escritorio:





Password: ITGAM_capcom12





6 Ambientación - Spring Boot

N/A

7 Ambientación - Angular

1. instalamos node o actualizamos con nvm

Instalar Node.js para gestión de dependencias (npm) y los comandos de Angular CLI

- Link de descarga
  - https://nodejs.org/es/download
- Configurar Variables de entorno
  - NPM_HOME =C:\Tools\node-v22.16.0-win-x64
  - PATH = %NPM_HOME %
- Validar la instalación
  - Abrir “Simbolo de Sistema” o CMD
  - y colocar el comando: node --version

```
MINGW64:/a/proyectos/angular/Abril
$ node --version
v20.20.0

mtjam@Armando MINGW64 /a/proyectos/angular/Abril
$ node -v
v20.20.0

mtjam@Armando MINGW64 /a/proyectos/angular/Abril
$ nvm install 22.16.0
Installation complete.
If you want to use this version, type:

nvm use 22.16.0

mtjam@Armando MINGW64 /a/proyectos/angular/Abril
$ nvm use 22.16.0
Now using node v22.16.0 (64-bit)

mtjam@Armando MINGW64 /a/proyectos/angular/Abril
$ node -v
v22.16.0
```

2. instalamos angular cli

- Link de Referencia
  - https://angular.dev/installation
- Ejecutar el siguiente comando: **npm install -g @angular/cli**
  - Abrir “Simbolo de Sistema” o CMD y colocar el comando.

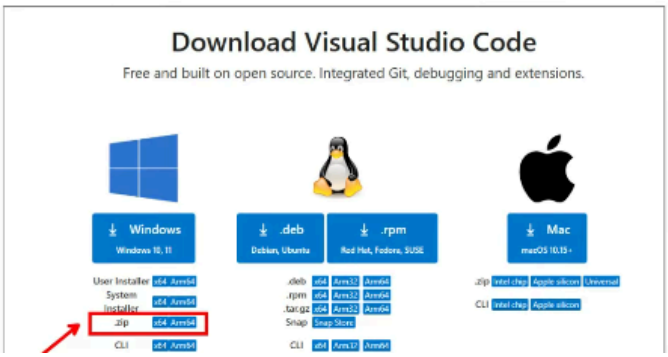
```
C:\Users\Perez>npm install -g @angular/cli

added 253 packages in 33s

49 packages are looking for funding
  run `npm fund` for details
```

```
C:\Users\Perez>ng --version
20.0.2
```

para editar el proyecto se ocupa visual studio code

5.4-Instalar Editor Visual Studio Code

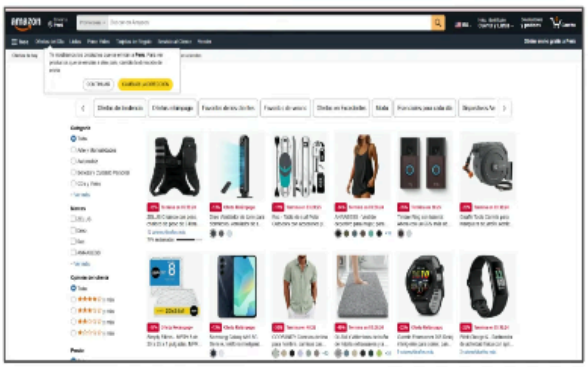
- Link de Referencia
  - https://code.visualstudio.com/Download
- Descargar en .zip



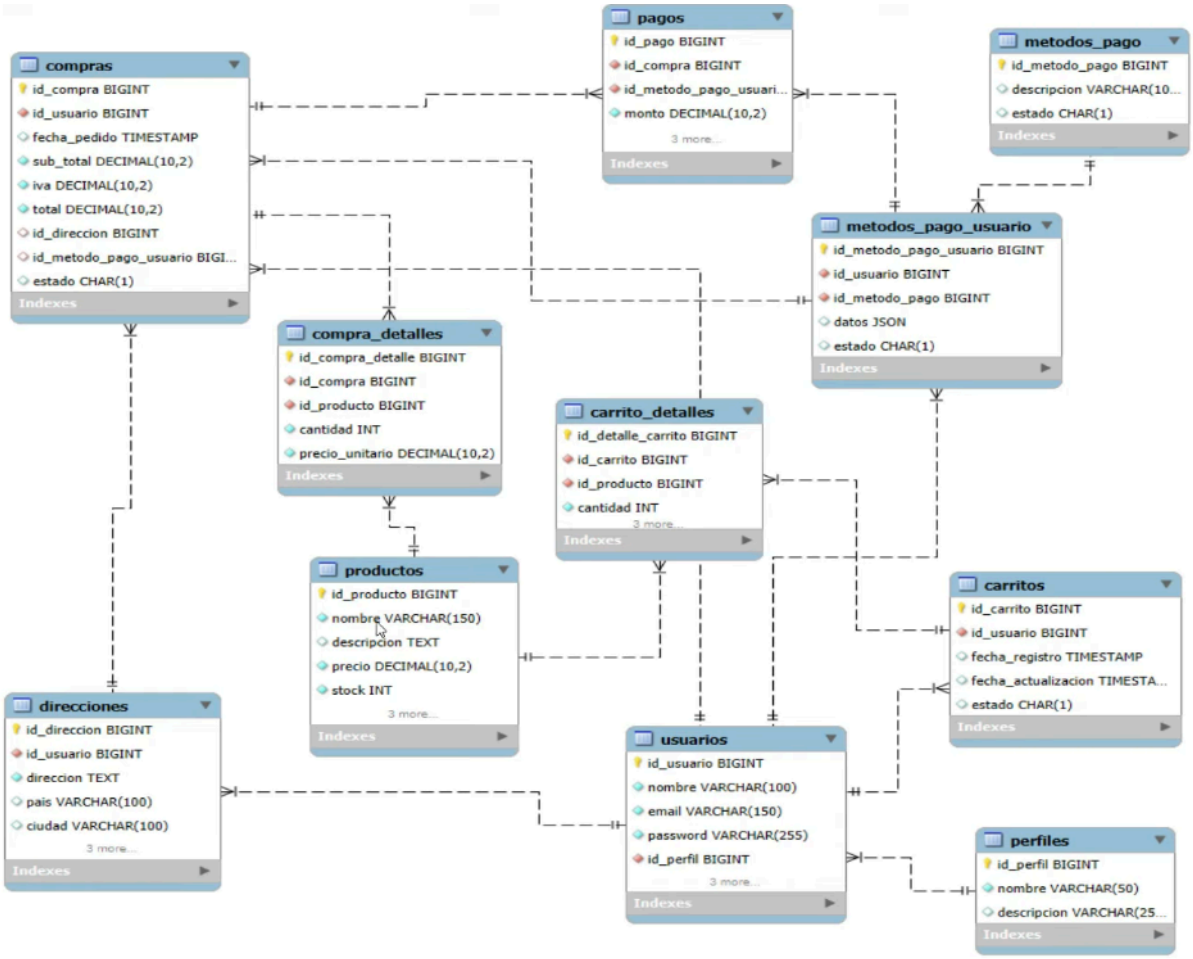
8 Actividades

1. Modelo Relacional de Base de datos

Cliente



Producto



2. Script de tablas en mysql

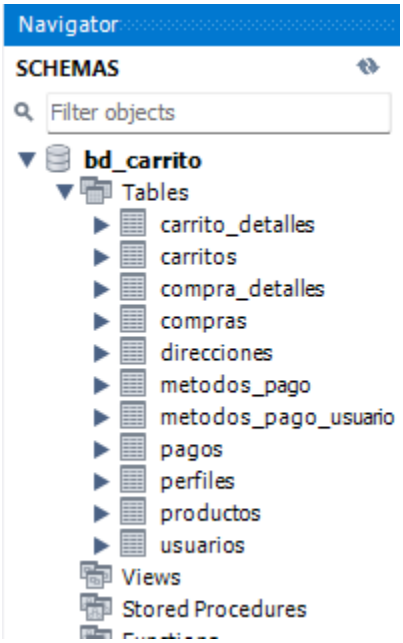
N	Nombre de tabla	llave primaria	llave secundaria	relación con
1	-- 0. Perfiles de Usuario CREATE TABLE perfiles (id_perfil BIGINT AUTO_INCREMENT PRIMARY KEY, nombre VARCHAR(50) NOT NULL UNIQUE, descripcion VARCHAR(255));	id_perfil		
2	-- 1. Usuarios	id_usuario		

	CREATE TABLE usuarios (id_usuario BIGINT AUTO_INCREMENT PRIMARY KEY, nombre VARCHAR(100) NOT NULL, email VARCHAR(150) UNIQUE NOT NULL, password VARCHAR(255) NOT NULL, id_perfil BIGINT NOT NULL, fecha_registro TIMESTAMP DEFAULT CURRENT_TIMESTAMP, fecha_actualizacion TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP, estado CHAR(1) DEFAULT 'A', FOREIGN KEY (id_perfil) REFERENCES perfiles(id_perfil));			
3	-- 2. Productos CREATE TABLE productos (id_producto BIGINT AUTO_INCREMENT PRIMARY KEY, nombre VARCHAR(150) NOT NULL, descripcion TEXT, precio DECIMAL(10,2) NOT NULL, stock INT NOT NULL, fecha_registro TIMESTAMP DEFAULT CURRENT_TIMESTAMP, fecha_actualizacion TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP, estado CHAR(1) DEFAULT 'A');	id_producto		
4	-- 3. Carritos CREATE TABLE carritos (id_carrito BIGINT AUTO_INCREMENT PRIMARY KEY, id_usuario BIGINT NOT NULL, fecha_registro TIMESTAMP DEFAULT CURRENT_TIMESTAMP, fecha_actualizacion TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP, estado CHAR(1) DEFAULT 'A',	id_carrito	id_usuario	usuarios

	FOREIGN KEY (id_usuario) REFERENCES usuarios(id_usuario));			
5	-- 4. Carrito Detalles CREATE TABLE carrito_detalles (id_detalle_carrito BIGINT AUTO_INCREMENT PRIMARY KEY, id_carrito BIGINT NOT NULL, id_producto BIGINT NOT NULL, cantidad INT NOT NULL, precio_unitario DECIMAL(10,2) NOT NULL, precio_igv DECIMAL(10,2) NOT NULL, precio_total DECIMAL(10,2) NOT NULL, FOREIGN KEY (id_carrito) REFERENCES carritos(id_carrito), FOREIGN KEY (id_producto) REFERENCES productos(id_producto));	id_detalle_carrito	id_carrito id_producto	carrito producto
6	-- 5. Direcciones CREATE TABLE direcciones (id_direccion BIGINT AUTO_INCREMENT PRIMARY KEY, id_usuario BIGINT NOT NULL, direccion TEXT NOT NULL, pais VARCHAR(100), ciudad VARCHAR(100), codigo_postal VARCHAR(20), es_predeterminada BOOLEAN DEFAULT FALSE, estado CHAR(1) DEFAULT 'A', FOREIGN KEY (id_usuario) REFERENCES usuarios(id_usuario));	id_direccion	id_usuario	usuario
7	-- 6. Métodos de Pago CREATE TABLE metodos_pago (id_metodo_pago BIGINT AUTO_INCREMENT PRIMARY KEY, descripcion VARCHAR(100) NULL, estado CHAR(1) DEFAULT 'A');	id_metodo_pago		
8	-- 7. Métodos de Pago de Usuarios	id_metodo_p	id_usuario	usuario

	CREATE TABLE metodos_pago_usuario (id_metodo_pago_usuario BIGINT AUTO_INCREMENT PRIMARY KEY, id_usuario BIGINT NOT NULL, id_metodo_pago BIGINT NOT NULL, datos JSON, estado CHAR(1) DEFAULT 'A', FOREIGN KEY (id_usuario) REFERENCES usuarios(id_usuario) ON DELETE CASCADE, FOREIGN KEY (id_metodo_pago) REFERENCES metodos_pago(id_metodo_pago));	ago_usuario	id_metodo_pago	metodos_pago
9	-- 8. Compras CREATE TABLE compras (id_compra BIGINT AUTO_INCREMENT PRIMARY KEY, id_usuario BIGINT NOT NULL, fecha_pedido TIMESTAMP DEFAULT CURRENT_TIMESTAMP, sub_total DECIMAL(10,2) NOT NULL, iva DECIMAL(10,2) NOT NULL, total DECIMAL(10,2) NOT NULL, id_direccion BIGINT, id_metodo_pago_usuario BIGINT, estado CHAR(1) DEFAULT 'A', FOREIGN KEY (id_usuario) REFERENCES usuarios(id_usuario), FOREIGN KEY (id_direccion) REFERENCES direcciones(id_direccion), FOREIGN KEY (id_metodo_pago_usuario) REFERENCES metodos_pago_usuario(id_metodo_pago_usuario));	id_compra	id_usuario id_direccion id_metodo_pago_usuario	usuario direcciones metodos_pago_usuario
10	-- 9. Compra Detalles CREATE TABLE compra_detalles (id_compra_detalle BIGINT AUTO_INCREMENT PRIMARY KEY, id_compra BIGINT NOT NULL, id_producto BIGINT NOT NULL, cantidad INT NOT NULL, precio_unitario DECIMAL(10,2) NOT NULL,	id_compra_detalle id_compra id_producto	id_compra id_producto	compras productos

	FOREIGN KEY (id_compra) REFERENCES compras(id_compra) ON DELETE CASCADE, FOREIGN KEY (id_producto) REFERENCES productos(id_producto));			
11	-- 10. Pagos CREATE TABLE pagos (id_pago BIGINT AUTO_INCREMENT PRIMARY KEY, id_compra BIGINT NOT NULL, id_metodo_pago_usuario BIGINT NOT NULL, monto DECIMAL(10,2) NOT NULL, fecha_pago TIMESTAMP DEFAULT CURRENT_TIMESTAMP, referencia_externa VARCHAR(255), estado CHAR(1) DEFAULT 'P', FOREIGN KEY (id_compra) REFERENCES compras(id_compra) ON DELETE CASCADE, FOREIGN KEY (id_metodo_pago_usuario) REFERENCES metodos_pago_usuario(id_metodo_pago_usuario));	id_pago	id_compra id_metodo_pago_usuario	compras metodos_pago_usuario



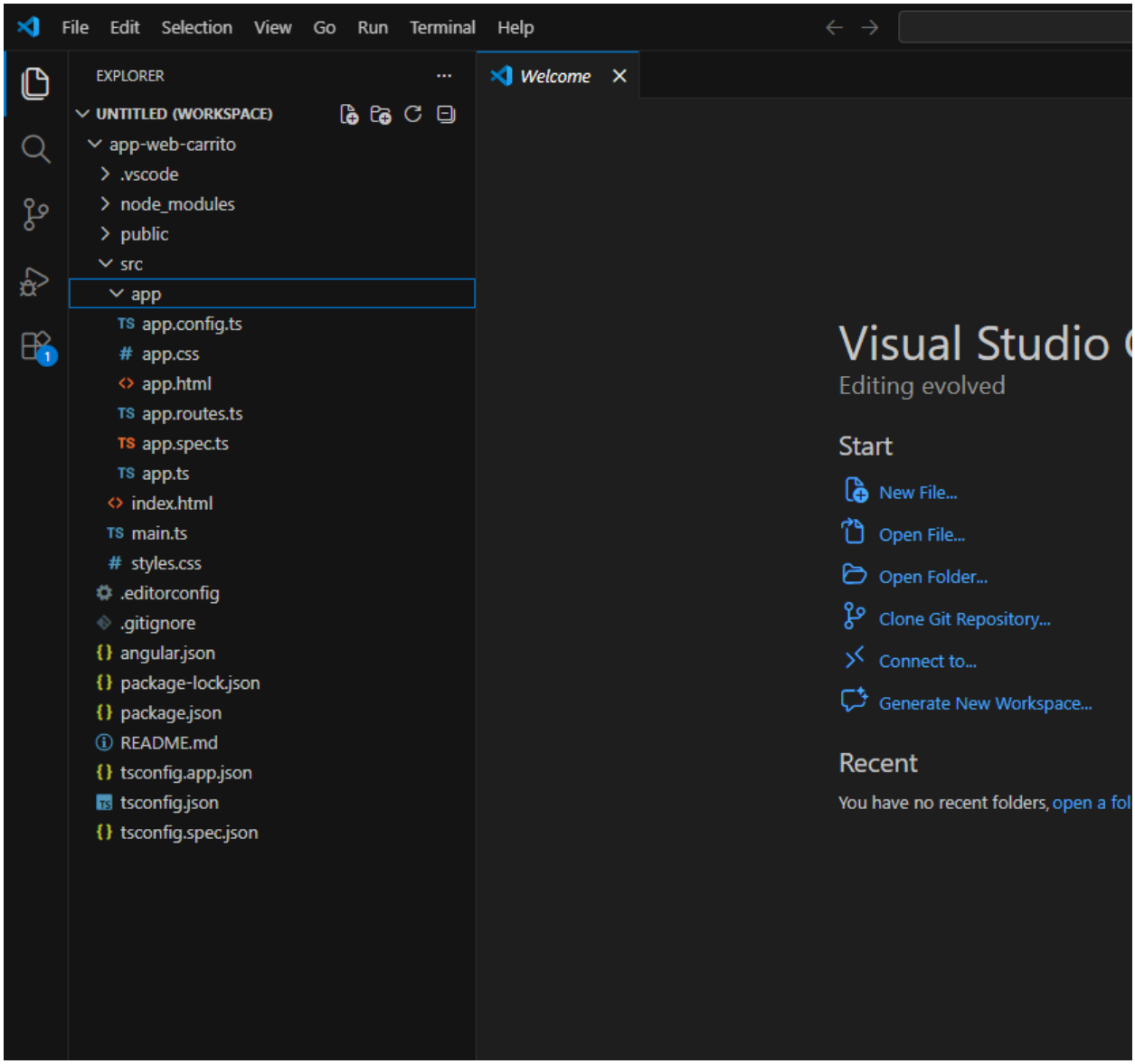
3. generamos el proyecto en angular

```
C:\Windows\System32\cmd.exe
Microsoft Windows [Versión 10.0.19045.5965]
(c) Microsoft Corporation. Todos los derechos reservados.

D:\SinFloo\1-Projects>ng new app-web-carrito
```

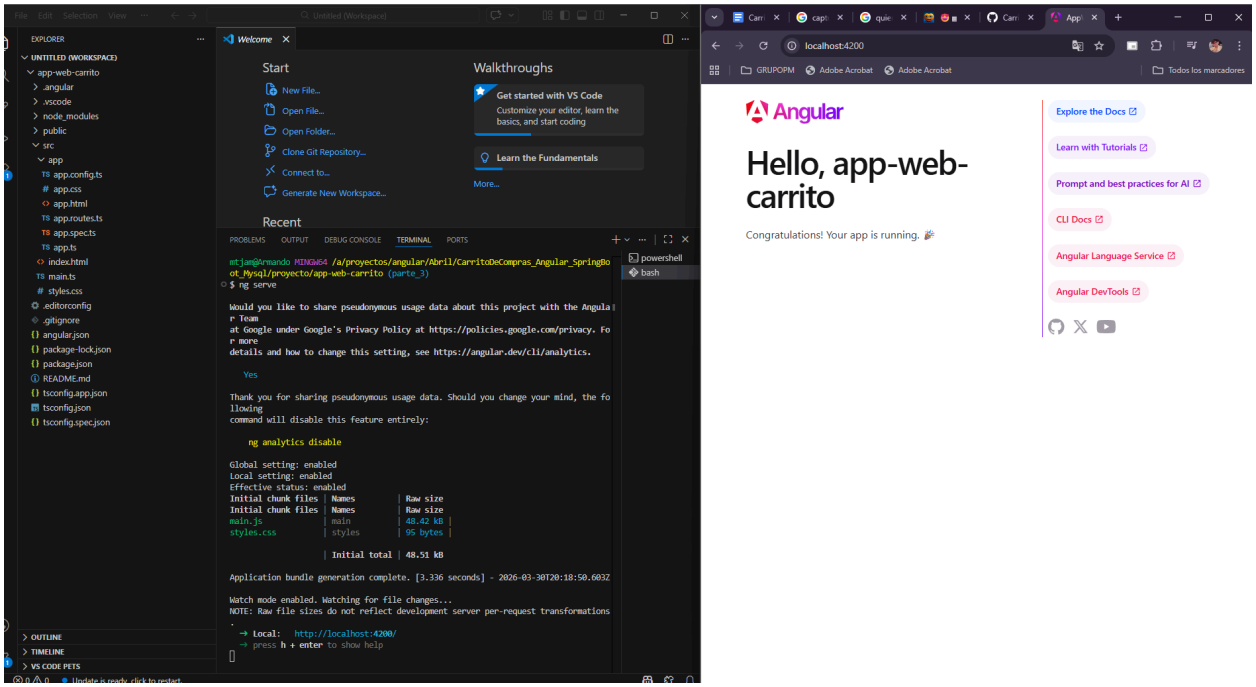
```
Microsoft Windows [Versión 10.0.19045.5965]
(c) Microsoft Corporation. Todos los derechos reservados.

D:\SinFloo\1-Projects>ng new app-web-carrito
√ Do you want to create a 'zoneless' application without zone.js (Developer Preview)? Yes
√ Which stylesheet format would you like to use? CSS              [ https://developer.mozilla.org/docs/Web/CSS
√ Do you want to enable Server-Side Rendering (SSR) and Static Site Generation (SSG/Prerendering)? No
CREATE app-web-carrito/angular.json (2343 bytes)
CREATE app-web-carrito/package.json (950 bytes)
CREATE app-web-carrito/README.md (1535 bytes)
CREATE app-web-carrito/tsconfig.json (1026 bytes)
CREATE app-web-carrito/.editorconfig (331 bytes)
CREATE app-web-carrito/.gitignore (629 bytes)
```

```
mtjam@Armando MINGW64 /a/proyectos/angular/Abril/CarritoDeCompras_Angular_SpringBoot_Mysql/proyecto (parte_3)
$ ng new app-web-carrito
√ Would you like to enable autocompletion? This will set up your terminal so pressing TAB while typing Angular CLI commands will show possible options and autocomplete arguments. (Enabling autocompletion will modify configuration files in your home directory.) Yes
Appended `source <(ng completion script)` to `C:\Users\mtjam\.bashrc`. Restart your terminal or run the following to autocomplete `ng` commands:

    source <(ng completion script)

Would you like to share pseudonymous usage data about this project with the Angular Team
at Google under Google's Privacy Policy at https://policies.google.com/privacy. For more
details and how to change this setting, see https://angular.dev/cli/analytics.

    Yes

Thank you for sharing pseudonymous usage data. Should you change your mind, the following
command will disable this feature entirely:

    ng analytics disable --global

Global setting: enabled
Local setting: No local workspace configuration file.
Effective status: enabled
√ Which stylesheet system would you like to use? CSS              [ https://developer.mozilla.org/docs/Web/CSS                    ]
√ Do you want to enable Server-Side Rendering (SSR) and Static Site Generation (SSG/Prerendering)? No
√ Which AI tools do you want to configure with Angular best practices? https://angular.dev/ai/develop-with-ai None
CREATE app-web-carrito/angular.json (2003 bytes)
CREATE app-web-carrito/package.json (982 bytes)
CREATE app-web-carrito/README.md (1525 bytes)
CREATE app-web-carrito/tsconfig.json (990 bytes)
CREATE app-web-carrito/.editorconfig (331 bytes)
CREATE app-web-carrito/.gitignore (666 bytes)
CREATE app-web-carrito/tsconfig.app.json (444 bytes)
CREATE app-web-carrito/tsconfig.spec.json (456 bytes)
CREATE app-web-carrito/.vscode/extensions.json (134 bytes)
CREATE app-web-carrito/.vscode/launch.json (490 bytes)
CREATE app-web-carrito/.vscode/mcp.json (188 bytes)
CREATE app-web-carrito/.vscode/tasks.json (1020 bytes)
CREATE app-web-carrito/src/main.ts (228 bytes)
CREATE app-web-carrito/src/index.html (312 bytes)
CREATE app-web-carrito/src/styles.css (81 bytes)
CREATE app-web-carrito/src/app/app.spec.ts (705 bytes)
CREATE app-web-carrito/src/app/app.ts (309 bytes)
CREATE app-web-carrito/src/app/app.css (0 bytes)
CREATE app-web-carrito/src/app/app.html (20448 bytes)
CREATE app-web-carrito/src/app/app.config.ts (324 bytes)
CREATE app-web-carrito/src/app/app.routes.ts (80 bytes)
CREATE app-web-carrito/public/favicon.ico (15086 bytes)
√ Packages installed successfully.
    Directory is already under version control. Skipping initialization of git.

mtjam@Armando MINGW64 /a/proyectos/angular/Abril/CarritoDeCompras_Angular_SpringBoot_Mysql/proyecto (parte_3)
$ dir
app-web-carrito

mtjam@Armando MINGW64 /a/proyectos/angular/Abril/CarritoDeCompras_Angular_SpringBoot_Mysql/proyecto (parte_3)
$ cd app-web-carrito/

mtjam@Armando MINGW64 /a/proyectos/angular/Abril/CarritoDeCompras_Angular_SpringBoot_Mysql/proyecto/app-web-carrito (parte_3)
$
```

4. Seleccionamos la carpeta del proyecto en angular con VSC



5. Ejecutamos el proyecto desde terminal en VSC con el comando ng serve


EXPLORER
UNTITLED (WORKSPACE)
app-web-carrito
.angular
.vscode
node_modules
public
src
app
app.config.ts
app.css
app.html
app.routes.ts
app.spec.ts
app.ts
index.html
main.ts
styles.css
.editorconfig
.gitignore
angular.json
package-lock.json
package.json
README.md
tsconfig.app.json
tsconfig.json
tsconfig.spec.json
OUTLINE
TIMELINE
VS CODE PETS
Welcome
Start
New File...
Open File...
Open Folder...
Clone Git Repository...
Connect to...
Generate New Workspace...
Walkthroughs
Get started with VS Code
Customize your editor, learn the basics, and start coding
Learn the Fundamentals
More...
Recent
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
mtjam@Armando MINGW64 /a/proyectos/angular/Abril/CarritoDeCompras_Angular_SpringBoot_Mysql/proyecto/app-web-carrito (parte_3)
$ ng serve
Would you like to share pseudonymous usage data about this project with the Angular Team
at Google under Google's Privacy Policy at https://policies.google.com/privacy. For more
details and how to change this setting, see https://angular.dev/cli/analytics.
Yes
Thank you for sharing pseudonymous usage data. Should you change your mind, the following
command will disable this feature entirely:
ng analytics disable
Global setting: enabled
Local setting: enabled
Effective status: enabled
Initial chunk files | Names | Raw size
Initial chunk files | Names | Raw size
main.js | main | 48.42 kB
styles.css | styles | 95 bytes
Initial total | 48.51 kB
Application bundle generation complete. [3.336 seconds] - 2026-03-30T20:18:50.603Z
Watch mode enabled. Watching for file changes...
NOTE: Raw file sizes do not reflect development server per-request transformations.
➜ Local: http://localhost:4200/
➜ press h + enter to show help
powershell
bash
localhost:4200
GRUPOPM
Adobe Acrobat
Adobe Acrobat
Todos los marcadores
Angular
Hello, app-web-carrito
Congratulations! Your app is running.
Explore the Docs
Learn with Tutorials
Prompt and best practices for AI
CLI Docs
Angular Language Service
Angular DevTools